

Deep Hedging vs. Black–Scholes Delta Hedging

A Personal Quant Finance Implementation Project

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¹Personal Project

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Abstract

This project report documents a personal implementation project in quantitative finance: building and comparing two option-hedging strategies – classical Black–Scholes delta hedging and a machine-learning-based “deep hedging” policy – under realistic transaction costs. The project was built independently in Python (NumPy/SciPy/Matplotlib only) and is inspired by and benchmarked against the methodology described in Suresh (2026) [1] and the deep hedging framework of Buehler et al. (2019) [3]. Beyond reproducing the core comparison, this project adds a transaction cost sensitivity sweep to study how the tradeoff between hedging cost and risk scales with market friction. Full source code is available on GitHub (link in Section 7).

1 Project Overview

The goal of this project was to answer a practical quant-finance question: *can a machine-learned hedging policy out-perform the classical Black–Scholes delta hedge once real trading frictions (transaction costs) are taken into account?* This required building, from scratch:

- a market simulator (Geometric Brownian Motion),
- a closed-form Black–Scholes pricing and delta-hedging engine,
- a small neural-network hedging policy trained without any deep learning framework, and
- an evaluation pipeline comparing both strategies on cost and risk.

This project was inspired by and follows the general methodology of Suresh (2026) [1], which compared these same two approaches in a simulated market. The implementation, code, and the transaction-cost sensitivity extension described below are my own independent work.

2 Methodology

2.1 Market Simulation

Stock price paths S_t were generated with Geometric Brownian Motion,

$$S_{t+\Delta t} = S_t \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right], \quad Z_t \sim \mathcal{N}(0, 1),$$

under risk-neutral drift, for pricing and hedging a European call option with strike K , risk-free rate r , volatility σ , and maturity T .

2.2 Black–Scholes Benchmark

The Black–Scholes call price and delta were computed in closed form at every rebalancing step. The hedge portfolio was rebalanced to hold Δ_t shares of the underlying at each step, with proportional transaction costs $c \cdot |\text{trade}| \cdot S_t$ deducted from the cash account, where c is the transaction cost rate.

2.3 Deep Hedging Policy

The learned hedging policy is a small feed-forward neural network (one hidden layer, four hidden units, tanh activation, sigmoid output) mapping the current state (moneyness, time-to-maturity) to a target hedge position in $[0, 1]$. Instead of using PyTorch or TensorFlow, the network was trained directly with `scipy.optimize` (L-BFGS-B, numerical gradients), minimizing mean squared terminal profit-and-loss (P&L) across simulated training paths, with transaction costs embedded directly into the P&L. This keeps the entire project dependency-light (NumPy, SciPy, Matplotlib) while preserving the core deep hedging idea: let the optimizer trade off hedge accuracy against trading cost.

2.4 Extension: Transaction Cost Sensitivity Sweep

As an extension beyond the base comparison, the deep hedge was retrained independently at transaction cost rates $c \in \{0.1\%, 0.5\%, 1\%, 2\%\}$ and compared against Black–Scholes at each level, to see how the size of the cost advantage and the size of the risk penalty change as trading frictions increase.

3 Results

3.1 Base Comparison

Using $S_0 = K = \$100$, $r = 2\%$, $\sigma = 20\%$, $T = 0.25$ years, 20 rebalancing steps, and a 1% transaction cost rate, 500 test paths produced the results in Table 1.

Metric	Black–Scholes	Deep Hedge
Mean cumulative transaction cost	1.959	1.439
Mean final P&L	−2.016	−1.450
P&L standard deviation	1.001	1.111

Table 1: Base comparison at a 1% transaction cost rate. The deep hedge cut transaction costs by roughly 26.5% relative to Black–Scholes, at the cost of higher P&L volatility – the same directional tradeoff reported by Suresh (2026), though the exact magnitude differs due to independent network architecture, training method, and simulation seed.

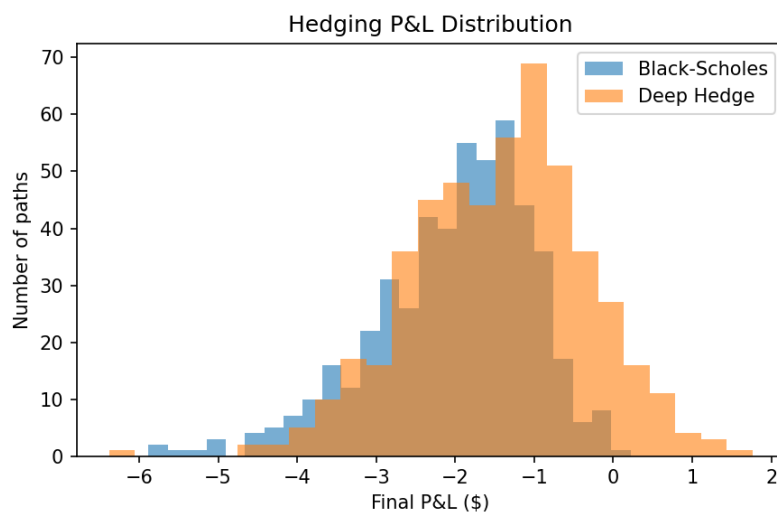


Figure 1: Final hedged P&L distribution across simulated paths. The Black–Scholes hedge is tighter; the deep hedge accepts more variance in exchange for lower trading costs.

3.2 Cost Sensitivity Sweep

Figure 2 shows the sweep across transaction cost rates. The deep hedge’s cost advantage grows with the cost rate – at $c = 0.1\%$ the two strategies cost about the same, while at $c = 2\%$ the deep hedge costs roughly 46% less than Black–Scholes. This indicates the learned policy only meaningfully departs from continuous delta hedging once frictions make doing so worthwhile.

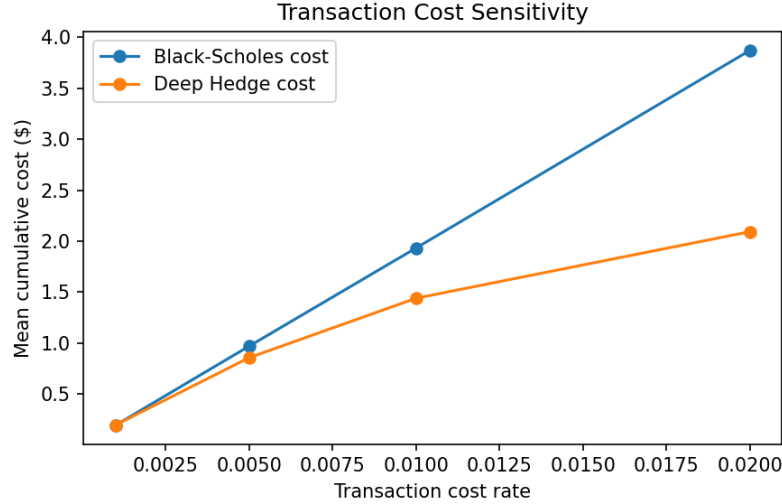


Figure 2: Mean cumulative transaction cost vs. transaction cost rate for both strategies.

4 Skills Demonstrated

- Monte Carlo / stochastic process simulation (GBM)
- Closed-form derivatives pricing (Black–Scholes)
- Numerical optimization for policy learning (gradient-free training of a neural network without a deep learning framework)
- Risk/cost tradeoff analysis and sensitivity testing
- Data visualization and results reporting
- Reproducible, dependency-light scientific Python code

5 Limitations

The market model is a simplified GBM with constant volatility, no jumps, and no bid-ask spread beyond the proportional cost term. The network is intentionally small and trained via numerical optimization rather than full backpropagation; a larger network trained end-to-end with a deep learning framework could produce different results. All results are based on simulated data and have not been validated against historical markets.

6 Conclusion

This project independently implemented and compared Black–Scholes delta hedging with a machine-learning-based deep hedging policy under transaction costs, confirming the directional tradeoff described in Suresh (2026): learned hedging can reduce trading costs at the expense of higher P&L volatility. The added transaction-cost sensitivity sweep shows this advantage grows with the size of market frictions. Natural next steps include a stochastic volatility model, an explicit risk-adjusted (e.g., CVaR) training objective, and backtesting on historical data.

7 Code Availability

Full source code, README, and setup instructions are available at: <https://github.com/YOUR-USERNAME/deep-hedging-project> (update this link once your repository is live).

References

- [1] Suresh, M. (2026). Deep Hedging Under Transaction Costs: A Comparison with Black–Scholes Delta Hedging. *SSRN Working Paper*.
- [2] Black, F., and Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3):637–654.
- [3] Buehler, H., Gonon, L., Teichmann, J., and Wood, B. (2019). Deep Hedging. *Quantitative Finance*, 19(8):1271–1291.